

# HVH Core Capabilities

## Rare and Specialty Disease

Dec. 20, 2016

# Agenda

1 Introductions

2 Unique Challenges

3 About Us

4 What We Do

5 Methodology

6 Team

Together, we can create a world  
where people with diseases  
don't have to find therapies...  
**THERAPIES FIND THEM.**



SEE EMILY'S STORY

**HMH**  
1PATIENT PRECISION ANALYTICS



Together, we can create a world  
where people with diseases  
don't have to find therapies...  
**THERAPIES FIND THEM.**

SEE EMILY'S STORY

# Stop the cycle of patient suffering

## Transforming Challenges:

- We can make a difference in patient's lives...but first we must find them
- Connecting patients to the care they need by accelerating the time to diagnosis
- We empower companies like yours to find people who are suffering from disease...  
*before they are even diagnosed*



# Providing the power and promise to find patients and shorten their journeys

a leading healthcare/  
biopharma marketing strategy  
& communications partner



a sophisticated  
analytics and predictive  
modeling leader



Leading the paradigm shift in biopharma commercial strategy to ensure that innovative therapies find the patients who desperately need them.

# Customized and collaborative consulting services

HVH is the only company that offers a variety of customized and collaborative consulting services, that are **proven to find patients in need**.



Revealing the unique identifiers of patients' journeys



New opportunities to find patients and accelerate diagnosis



Providing truly actionable insights of undiagnosed patients



# Revealing the unique identifiers of patients' journeys



Most patient journey analyses begin at the point of diagnosis, until now. Our analytics process gives you the ability to deeply understand where patients have been and predict where they will be:

- Quantify which patient characteristics are leading predictors of disease 6, 12, and 18 months before diagnosis
- Create a True treatment algorithm based on real-world evidence, not based on few KOL opinions
- Reveal unique identifiers of diagnosed patients' journey, beyond what is currently known
- Understand when and why drug switching occurs



# Uncovering new opportunities to find patients and accelerate diagnosis



We find patients for clinical trial recruitment or to expand your market share

- Provide ranked lists of HCPs who are treating patients who most likely have the disease
- Deliver HIPAA compliant, de-identified actual patient targets and journeys
- Locate patients who may have fallen out of the system



# Providing truly actionable insights of undiagnosed patients



We offer a real-world-evidence based approach that surpasses what is currently available

- Create a clear definition of the diagnosed patient for indications lacking specific ICD-9 or ICD-10 codes.
- Validate and compare the market size of diagnosed patients between real world data and published literature
- Provide a deeper understanding of market share, competitive assessment, and therapeutic approaches to disease management



# Providing truly actionable insights of undiagnosed patients



We offer a real-world-evidence based approach that surpasses what is currently available

- Create a clear definition of the diagnosed patient for indications lacking specific ICD-9 or ICD-10 codes.
- Validate and compare the market size of diagnosed patients between real world data and published literature
- Provide a deeper understanding of market share, competitive assessment, and therapeutic approaches to disease management

# Providing truly actionable insights of undiagnosed patients

We offer a real-world-evidence based approach that surpasses what is currently available

- Create a clear definition of the diagnosed patient for indications lacking specific ICD-9 or ICD-10 codes.
- Validate and compare the market size of diagnosed patients between real world data and published literature
- Provide a deeper understanding of market share, competitive assessment, and therapeutic approaches to disease management





# Providing truly actionable insights of undiagnosed patients

We offer a real-world-evidence based approach that surpasses what is currently available

- Create a clear definition of the diagnosed patient for indications lacking specific ICD-9 or ICD-10 codes.
- Validate and compare the market size of diagnosed patients between real world data and published literature
- Provide a deeper understanding of market share, competitive assessment, and therapeutic approaches to disease management

# Target HCPs with undiagnosed, drug naive patients

## Situation

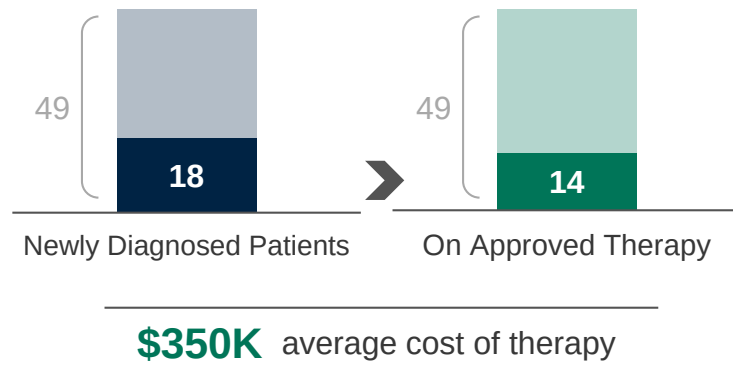
HVH engaged a biotech company with a commercialized product in the ultra orphan disease space, when there was only 1 other competitor product. The commercial strategy was to capture market share but 5 years post-launch, their growth slowed and the strategy shifted to market expansion and finding diagnosed or undiagnosed drug naive patients.

## Approach

HVH leveraged their proprietary platform and applied predictive modeling against large national claims database to deliver list of HCPs names with NPI numbers & addresses. Client deployed their sales team, to activate disease education messaging those HCPs who may have had undiagnosed patients.

## Results

At the end of 1 year, client was able to confirm 18 out of 49 were newly diagnosed patients and 14 of those were confirmed to be on an approved therapy. The average cost of the therapy was \$350K resulting in significant revenue capture for the client.



# Connecting you to patients via a patent-pending, predictive modeling 3-step process

1

## PATIENT DEFINITION

Patients are defined by analyzing clinical and natural history data, expert input, and drug and diagnosis codes over time

2

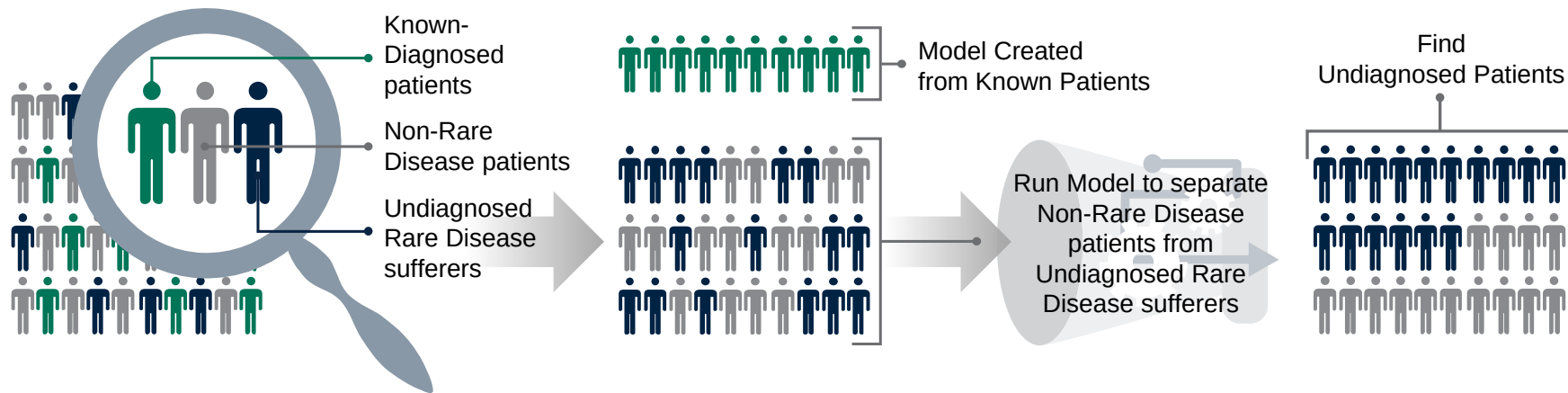
## MODEL CREATION

Disease model creation is achieved by refining the patient profile and incorporating machine learning

3

## PREDICTION

Once a disease model is developed, we then apply that to the remaining population which enables us to identify undiagnosed patients



# HVH's methodology validated in peer-reviewed publication



## Predictive Analytics: A Case Study in Machine-Learning and Claims Databases

David A. Kvanz, MS, RPh, FASHP; Marcus N. Sredzinski, PharmD; and Celynda G. Tadlock, PharmD, MBA

### ABSTRACT

The healthcare industry generates massive collections of data—big data—with the potential to reveal insights into optimizing costs and outcomes if analyzed with the proper tools. Machine learning and predictive algorithms are already in common use in other fields. In healthcare, the largest datasets with the broadest relevance to the US population may reside in claims databases. Analyzing such databases with the latest tools may find rare or hard-to-diagnose diseases that needlessly consume healthcare dollars before proper diagnoses are made.

**Objective:** This study focused the power of modern analytics on hereditary angioedema (HAE), a single rare disease, because it exhibits features of diseases associated with high costs: rare, hard to diagnose, progressive, and takes a long time from diagnosis to appropriate treatment. Despite the availability of effective therapies, misdiagnoses and underdiagnoses of HAE result in significant burden to the healthcare system.

**Methods:** A 3-stage process was applied to a claims database to: a) define the characteristics (diagnoses, procedures, therapies, and providers) of patients in the database already being treated for HAE, b) use those characteristics to create a model of patients with HAE, and c) use the model to identify patients with HAE in the database who were not yet diagnosed.

**Conclusions:** This study successfully demonstrated the ability of this state-of-the-art predictive analysis to find rare-disease patients in a large and complex database. This information could be valuable to claims managers and employers who may realize savings by helping physicians bring these patients to appropriate treatment sooner.

*Am J Pharm Benefits.* 2016;8(6):214-219



“

The study demonstrated the ability of this state-of-the-art predictive analysis to find rare disease patients in a large and complex database, **shortening the time to diagnosis** and relieving patients of the frustration of going from doctor to doctor without a confirmed diagnosis.

”



# Stage 1: Patient Definition Process

## PURPOSE:

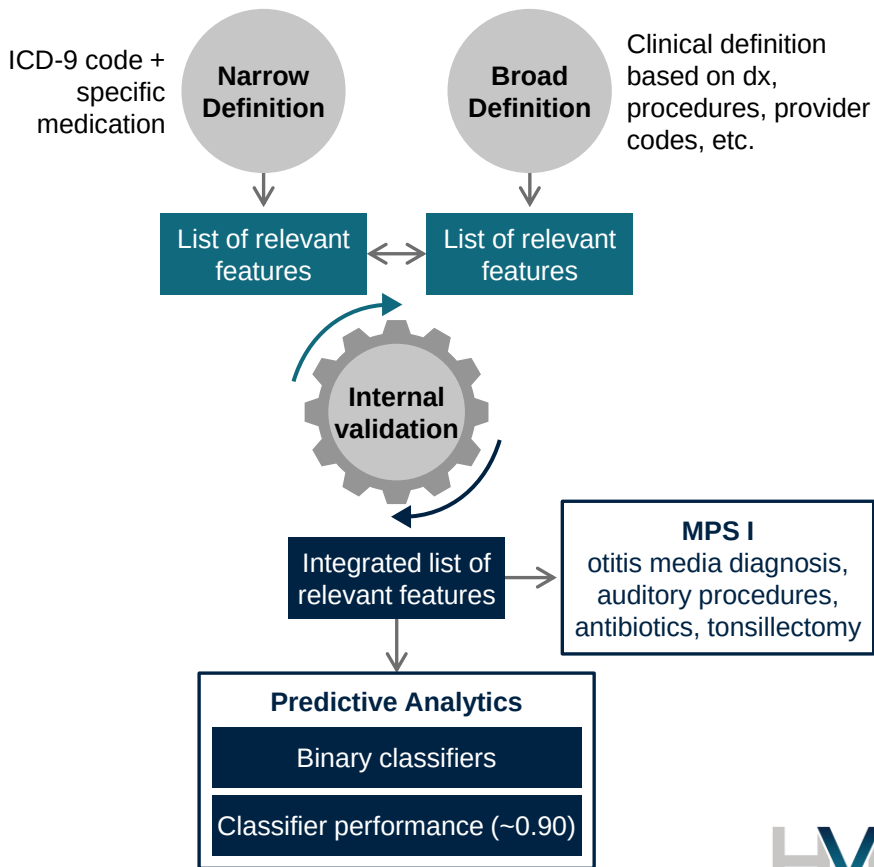
Define the foundational patient population upon which to build the predictive analytics solution

## METHOD:

- (1) Extract discriminative features spanning procedures, diagnoses, providers and drug codes
- (2) Develop binary classifiers to separate normal and undiagnosed patients

## RESULTS:

- Successful internal validation – resulting features consistent with known clinical features of disease
- Predictive model performance (measured by Area Under the Curve (AUC)) ranges between 0.86 – 0.92 indicating strong capability to separate the undiagnosed patient population



# Stage 2: Model Creation Process

## FEATURE SELECTION

### FEATURE SELECTION and EXTRACTION

- Extract data elements from patient definition
- Find characteristics that separate patient populations
- Review features generated for clinical validity

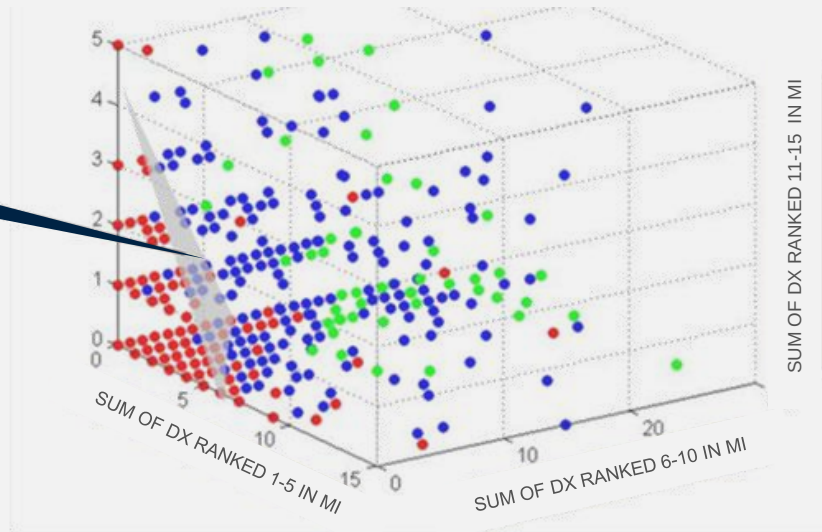
### MODEL DEVELOPMENT

- Select models and testing, training & validation strategies
  - Supervised: SVM
  - Unsupervised: Clustering
- Learning Using Privileged Information
- Generate training, testing & validation data samples
- Optimize model performance

3-D projection where axes are sums of frequencies of diagnosis groups among top 15 diagnoses

● NON-POMPE PATIENT ● POMPE PATIENT ● POMPE SUSPECT

Decision Rule



# Stage 2: Model Creation Process

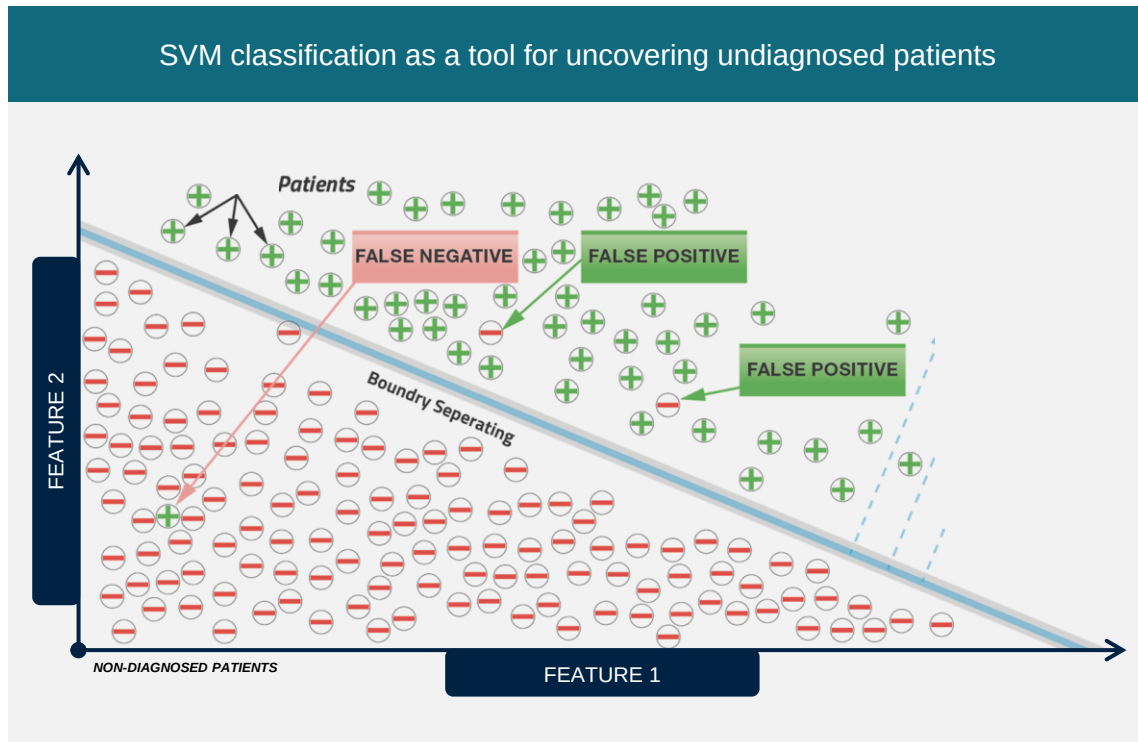
## CLASSIFIER DEVELOPMENT

### UNDIAGNOSED PATIENT CLASSIFIER

- Two classes assigned
  - True positive diagnosis
  - True negative diagnosis
- Rank patients based on model score

### UNDIAGNOSED PATIENT PREDICTION

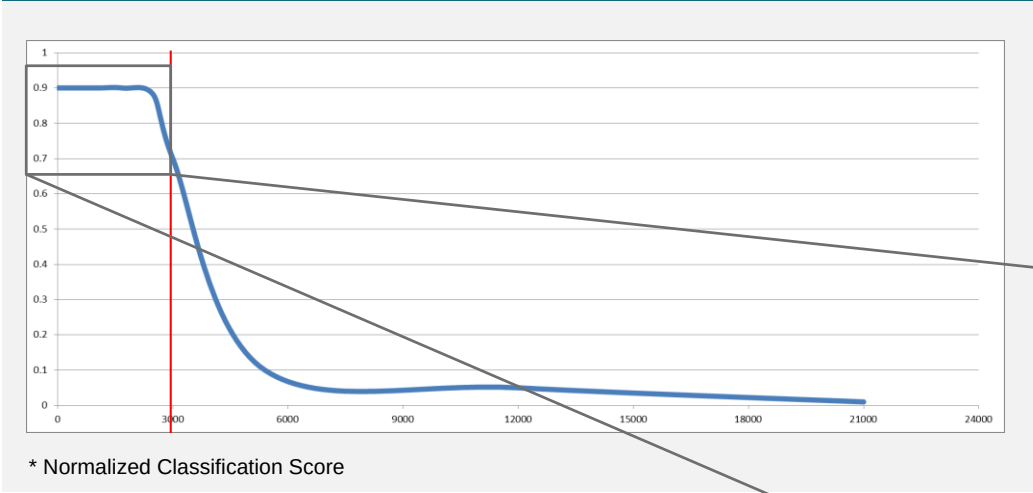
- Calculate features for each patient in claims database
- Input individual patient features into classification model
- Rank classifier output and generate candidate patient list
- Extract demographics and provider information



# Stage 3: Prediction Process:

## PRELIMINARY PATIENT LIST

~3,000 patients >0.7\* (Disease “X”)

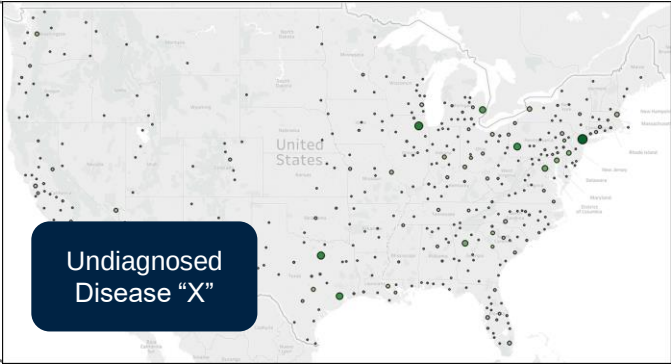


XXXXXX

- XXXXX
- XXXX

| Patient # | Gender | Age | MSA        | HCP | NPI | Address |
|-----------|--------|-----|------------|-----|-----|---------|
| 1         | Male   | 8   | Sacramento |     |     |         |
| 2         | Female | 14  | Charlotte  |     |     |         |
| 3         | Male   | 4   | Atlanta    |     |     |         |

Will be included in final report





# Multidisciplinary team

Our multi-disciplinary team is comprised of experts in subject matter, strategy, and analytics/data science who will bring years of experience to your project



**Jeff Ceitlin**  
Chief Commercial Officer  
HVH



**Tara Grabowsky, MD**  
Chief Medical Officer  
HVH



**Oodaye Shukla**  
Chief Data Scientist  
HVH



**Rob Fair**  
Operations Director  
HVH

Specific bios located in the appendix.

# Summary

|                                                                        | HMH | Others, including internal teams |
|------------------------------------------------------------------------|-----|----------------------------------|
| Proven pull-through patient-finding with 95% confidence interval       | ✓   |                                  |
| Patented technology                                                    | ✓   |                                  |
| Validated methodology (source: HAE paper TBD)                          | ✓   |                                  |
| 40+ years of experience in predictive modeling                         | ✓   |                                  |
| Multi-disciplinary approach to deliverable (medical, data, & strategy) |     |                                  |

## Example Project Deliverables

## Patient Definition

- Create “true” definition of chronic migraine patient
- Examine key events in medical history
- Patient definition becomes the basis of all analysis

## Diagnosed Patient Characteristics

- o Identify unique characteristics to message patients and HCPs effectively in time periods before diagnosis (6 months, 12 months and/or 18 months)

## Undiagnosed Patient Characteristics

- Diagnosed characteristics to find undiagnosed patients
- Identify unique characteristics to message patients and HCPs effectively

## Optimized Patient Lists

- o Patient candidates containing:  
Metropolitan statistical area  
(MSA), associated provider name  
(NPI # / address) and patient  
demographics (gender, age)

